

Definition 1.12. A *local ring* is a commutative ring with a unique maximal ideal.

Definition 1.13. The *residue field* of a local ring A with maximal ideal $\mathfrak{m}$ is the field $A/\mathfrak{m}$.

We can now see how to determine the valuation v corresponding to a discrete valuation ring A . Given a discrete valuation ring A with unique maximal ideal $\mathfrak{m}$, we may define $v: A \rightarrow \mathbb{Z}$ by letting $v(a)$ be the unique integer n for which $(a) = \mathfrak{m}^n$ and $v(0) := \infty$. Extending v to the fraction field k of A via $v(a/b) := v(a) - v(b)$ gives a discrete valuation v on k for which $A = \{x \in k : v(x) \geq 0\}$ is the corresponding valuation ring.

Notice that any discrete valuation v on k with A as its valuation ring must satisfy $v(\pi) = 1$ for some $\pi \in \mathfrak{m}$ (otherwise $v(k) \neq \mathbb{Z}$), and we then have $v(\pi) = 1$ if and only if $\mathfrak{m} = (\pi)$. Moreover, v must then coincide with the discrete valuation we just defined: for any DVR A , the discrete valuation on the fraction field of A that yields A as its valuation ring is uniquely determined. It follows that we could have defined a uniformizer to be any generator of the maximal ideal of A without reference to a valuation.

Example 1.14. For the p -adic valuation $v_p: \mathbb{Q} \rightarrow \mathbb{Z} \cup \{\infty\}$ we have the valuation ring

$$\mathbb{Z}_{(p)} := \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, p \nmid b \right\},$$

with maximal ideal $\mathfrak{m} = (p)$; this is the *localization* of the ring $\mathbb{Z}$ at the prime ideal (p) . The residue field is $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \simeq \mathbb{Z}/p\mathbb{Z} \simeq \mathbb{F}_p$.

Example 1.15. For any field k , the valuation $v: k((t)) \rightarrow \mathbb{Z} \cup \{\infty\}$ on the field of Laurent series over k defined by

$$v \left(\sum_{n \geq n_0} a_n t^n \right) = n_0,$$

where $a_{n_0} \neq 0$, has valuation ring $k[[t]]$, the power series ring over k . For $f \in k((t))^\times$, the valuation $v(f) \in \mathbb{Z}$ is the *order of vanishing* of f at zero. For every $\alpha \in k$ one can similarly define a valuation v_α on k as the order of vanishing of f at α by taking the Laurent series expansion of f about α .

1.5 Discrete Valuation Rings

Discrete valuation rings are in many respects the nicest rings that are not fields. In addition to being an integral domain, every discrete valuation ring A enjoys the following properties:

- *noetherian*: Every increasing sequence $I_1 \subseteq I_2 \subseteq \cdots$ of ideals eventually stabilizes; equivalently, every ideal is finitely generated.
- *principal ideal domain*: Every ideal is principal (generated by a single element).
- *local*: There is a unique maximal ideal $\mathfrak{m}$.
- *dimension one*: The (Krull) *dimension* of a ring R is the supremum of the lengths n of all chains of prime ideals $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ (which need not be finite, in general). For DVRs, $(0) \subseteq \mathfrak{m}$ is the longest chain of prime ideals, with length 1.
- *regular*: The dimension of the $A/\mathfrak{m}$ -vector space $\mathfrak{m}/\mathfrak{m}^2$ is equal to the dimension of A . Non-local rings are regular if this holds for every localization at a prime ideal.
- *integrally closed* (or *normal*): Every element of the fraction field of A that is the root of a monic polynomial in $A[x]$ lies in A .